

EX. NO: 1 — CAESAR CIPHER

Aim:

To implement the Caesar Cipher substitution technique using C.

Algorithm:

1. Read plaintext.
2. Read key.
3. Shift each character by the key.
4. Apply modulo 26.
5. Display encrypted and decrypted text.

Program:

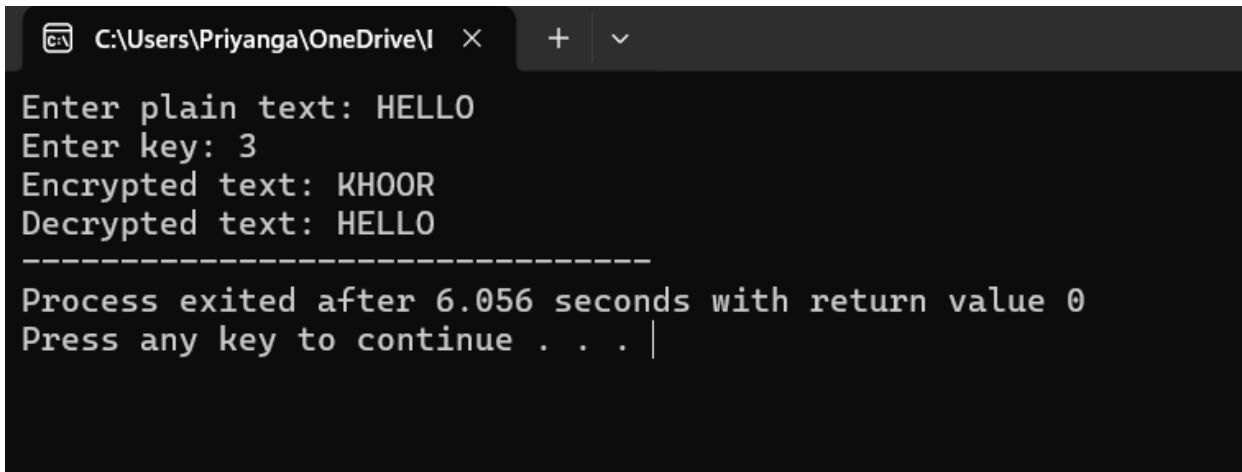
```
#include <stdio.h>
#include <ctype.h>
int main()
{
    char text[100], cipher[100], decrypt[100];
    int key, i;
    printf("Enter plain text: ");
    scanf("%s", text);
    printf("Enter key: ");
    scanf("%d", &key);
    /* Encryption */
    for(i = 0; text[i] != '\0'; i++)
    {
        if(isupper(text[i]))
            cipher[i] = (text[i] - 'A' + key) % 26 + 'A';
        else
            cipher[i] = (text[i] - 'a' + key) % 26 + 'a';
    }
```

```

cipher[i] = '\0';
printf("Encrypted text: %s", cipher);
/* Decryption */
for(i = 0; cipher[i] != '\0'; i++)
{
    if(isupper(cipher[i]))
        decrypt[i] = (cipher[i] - 'A' - key + 26) % 26 + 'A';
    else
        decrypt[i] = (cipher[i] - 'a' - key + 26) % 26 + 'a';
}
decrypt[i] = '\0';
printf("\nDecrypted text: %s", decrypt);
return 0;
}

```

Output:



```

C:\Users\Priyanga\OneDrive\I >
Enter plain text: HELLO
Enter key: 3
Encrypted text: KHOOR
Decrypted text: HELLO
-----
Process exited after 6.056 seconds with return value 0
Press any key to continue . . . |

```

Result:

Thus, the Caesar Cipher substitution technique was implemented successfully using C.